Ethics

JOHNS HOPKINS
WHITING SCHOOL
of ENGINEERING

Overview

- Introduction and initial examples
- Stepping through NLP pipeline
- Course things

- Scope:
  - Not a formal overview of ethics frameworks, an overview of ethical challenges in NLP, driven by examples
  - What are some of the ethical challenges in this space? Why are they difficult? Why should you care about them?

How can we develop AI (or NLP) for good and not for bad?

Decisions we make about our data, methods, and tools are tied up with their impact on people and societies

Example: Are there some applications we should not build?

Hypothetical case: should we build a classifier to predict someone's sexual orientation from their photo?

Why might we want to do this?

Sexual Orientation Classifier

Who can be harmed by such a classifier?

- Personal attributes (gender, race, sexual orientation, religion) are complex social constructs, not categorial/binary, are dynamic, are private and often not visible publicly
- These are properties for which people are often discriminated against
  - In many places being gay is prosecutable
  - Such a classifier might affect people's employment, family relationships, health care opportunities, etc.

Additional Ethical Questions

- Who can benefit from such a classifier?
- Where does the training data come from?
- Did anyone consent?

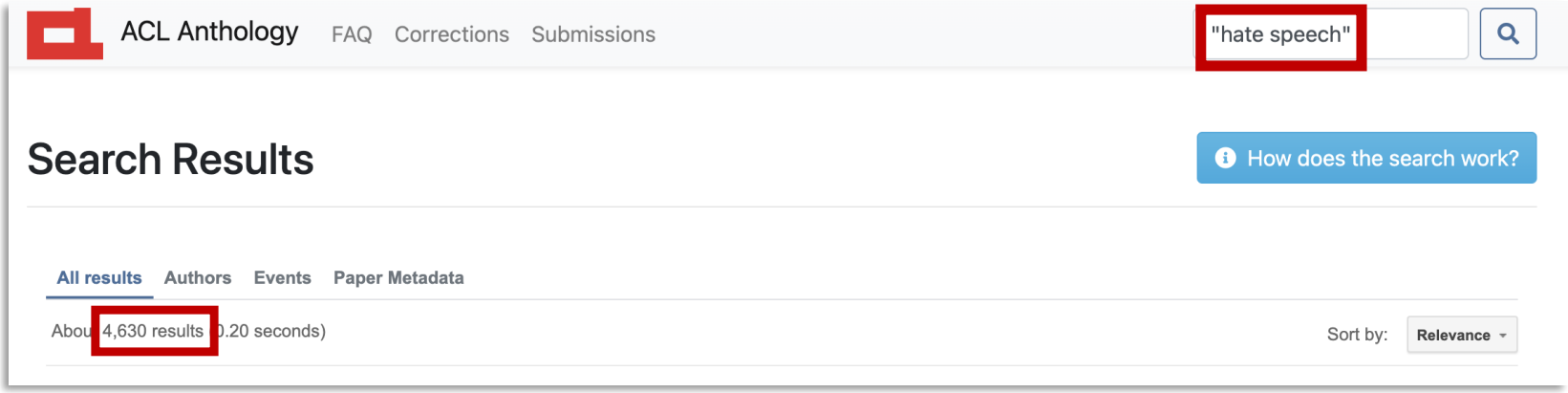
Most examples are not so straightforward

Problem:

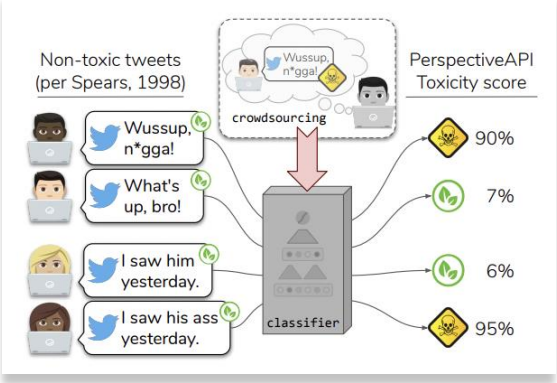
- Hate speech and offensive language are prevalent on the internet and can lead to tangible harms
- Marginalized people are disproportionally targets of hate speech
- Manually identifying hate speech is difficult for human moderators
  - Too much volume to keep up with
  - Mental toll of reading offensive content

Technical Solution

- Build NLP models to identify hate speech automatically



More Problems: NLP models are biased



[Sap et al. 2019]

Recall:

- Annotators tend to mis-label AAE as toxic or offensive
- This leads to models misclassifying AAE speech

More Problems: NLP models are biased

Term	Toxic	Overall
atheist	0.09%	0.10%
queer	0.30%	0.06%
gay	3%	0.50%
transgender	0.04%	0.02%
lesbian	0.10%	0.04%
homosexual	0.80%	0.20%
feminist	0.05%	0.05%
black	0.70%	0.60%
white	0.90%	0.70%
heterosexual	0.02%	0.03%
islam	0.10%	0.08%
muslim	0.20%	0.10%
bisexual	0.01%	0.03%

Table 1: Frequency of identity terms in toxic comments and overall.

[Dixon et al. 2018]

- Related work has shown data with identity terms is also likely to mis-labeled as toxic/offensive by models
- If we deploy these systems, they potentially end up censoring the very people they are supposedly protecting

Even more problems

- How do define what is offensive or hate speech?
  - Norms differ widely in different communities
  - Setting a universal standard is enforcing a majority viewpoint
- Who has control of the technology?
  - Concentrating power in few hands
- How might this technology be abused? (dual use potential)
  - Hate speech generator
  - Censorship

Solutions?

- Don’t build NLP for hate speech detection?
- But then what about all the hate speech on the internet?
- Maybe we should ban social media? The internet?

Stepping through the NLP Pipeline

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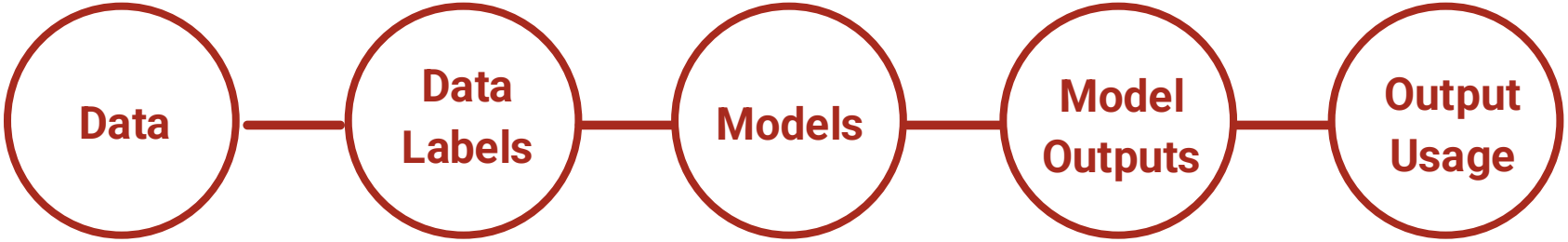
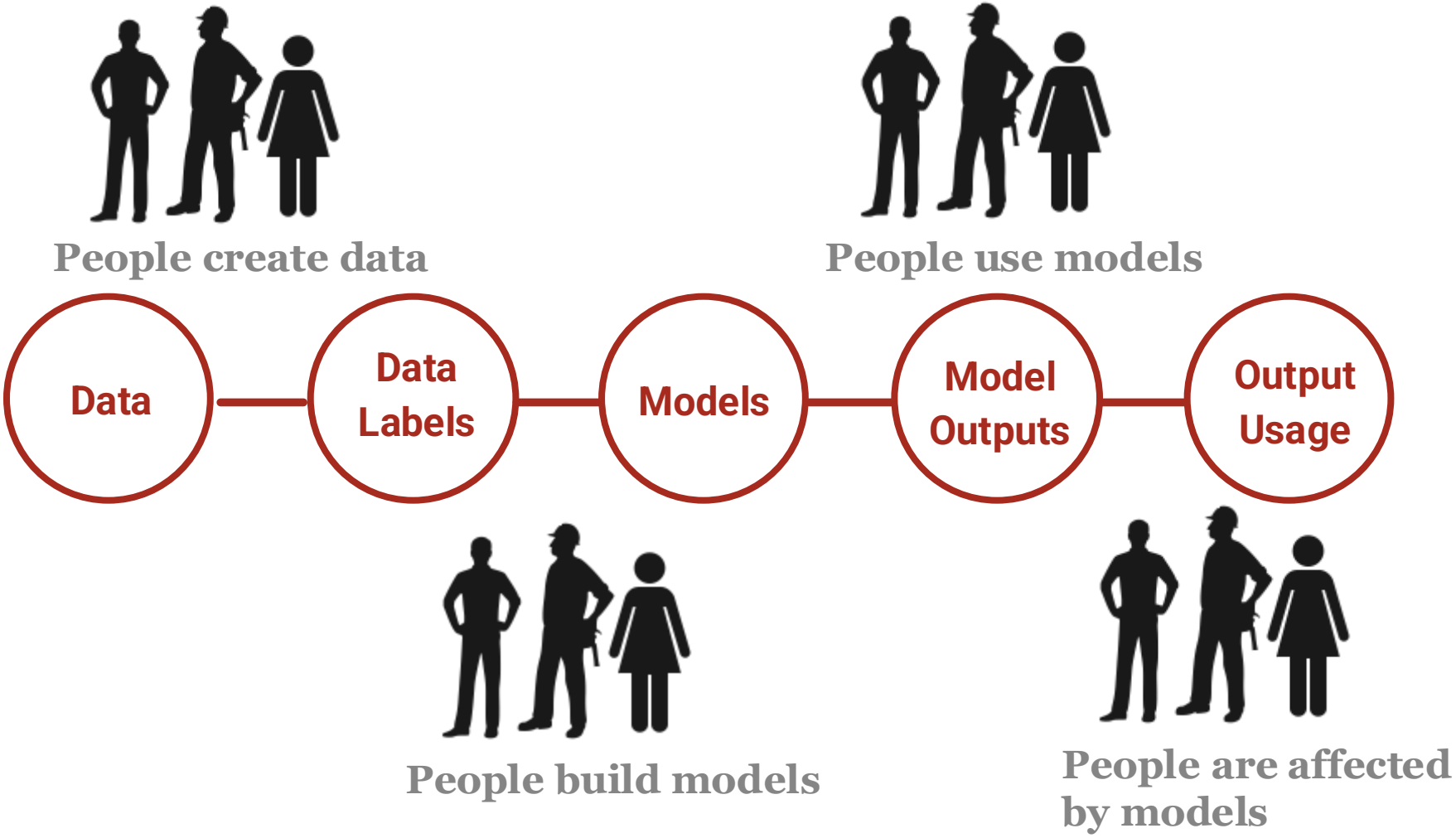
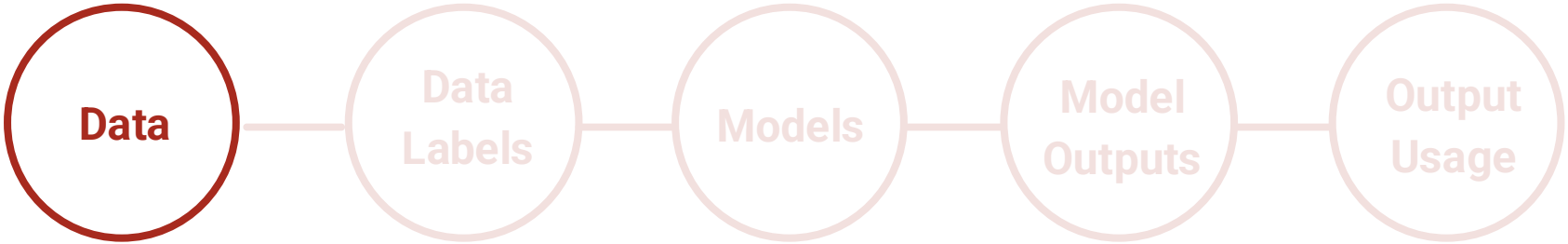
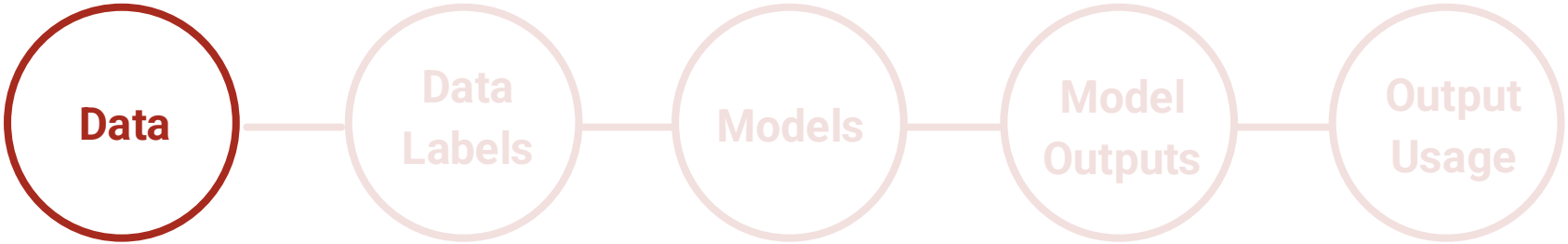

Data
Data Labels
Models
Model Outputs
Output Usage


People create data
People use models
Data
Data Labels
Models
Model Outputs
Output Usage
People build models
People are affected by models



- Bias and Representation
  - Models are prone to absorbing and amplifying data biases
  - Will our model assume that all doctors are men and all nurses are women?
- Ownership and copyright
  - Who owns the data?
  - Did we have permission to collect this data?
  - Did individuals consent to this use of their data?
- Privacy
  - Models are prone to memorizing and outputting sensitive information from data



- An example:
  - Collection data from Twitter

- What are some considerations?
  - Platform terms of service
  - Users may have posted data publicly, but they didn't explicitly consent to this analysis

What do Twitter users think of the use of their data?

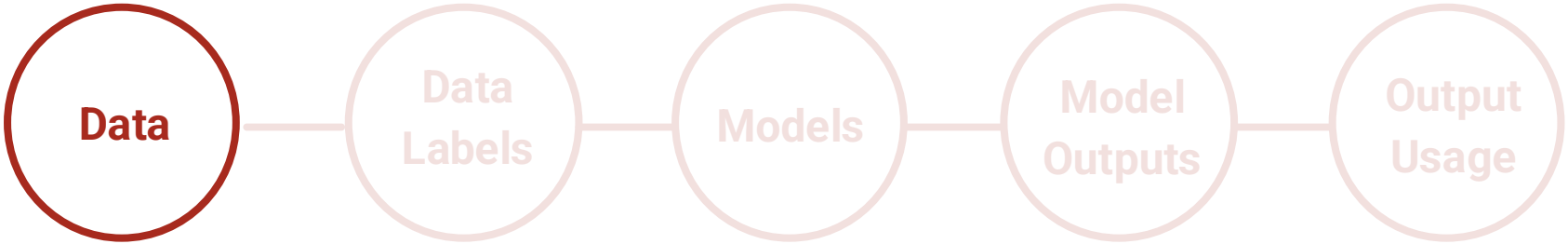
- Methods: recruit survey participants through Amazon Mechanical Turk, selecting for people who use Twitter: 368 total respondents

- Most (61.2%) users were not aware that their public tweets could be used by researchers. 42.7% thought researchers were not allowed to use tweets without user's consent (because of Twitter TOS, copyright law, or research ethics)

- The majority felt that researchers should not be able to use tweets without consent

- Attitudes are highly contextual, depending on factors such as how the research is conducted or disseminated, who is conducting it, and what the study is about

Fiesler, C., & Proferes, N. (2018). "Participant" Perceptions of Twitter Research Ethics. *Social Media + Society*, *4*(1). https://doi.org/10.1177/2056305118763366 (Original work published 2018)



- An example:
  - Collection data from Twitter

- What are some considerations?
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  - Users may have posted data publicly, but they didn't explicitly consent to this analysis
  - What if someone deletes their tweet after it's been collected for research?

Informed Consent for working with deleted tweets

- Study goals: examine when social media users correct rumors

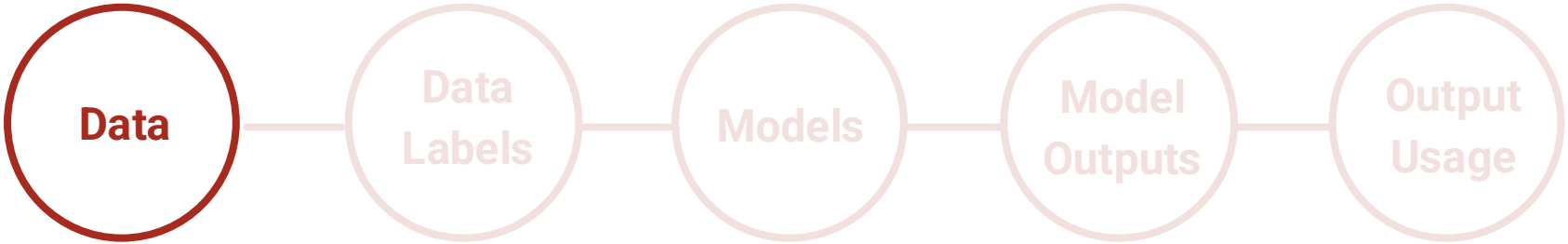
- Methods:
  - Live collect data
  - Identify when tweets are deleted
  - Reach out to users who deleted tweets to participate in the study

- **Remove deleted data except from users who consented to participate in the study**

Ethical Considerations: Identifying Deleted Tweets and Interviewing Rumor Tweeters

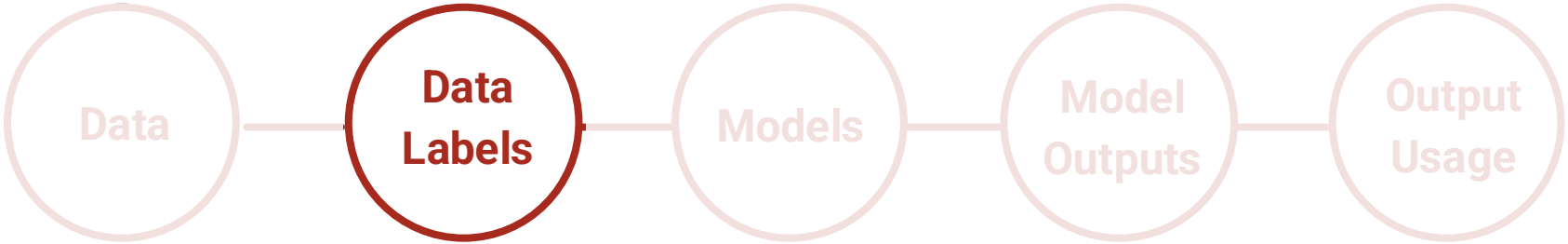
We encountered significant ethical challenges around working with deleted tweets as well as recruiting, interviewing and reporting results from online users who participated in a socially stigmatized activity: passing along rumors. For the deleted tweets, we followed the protocol outlined in [25]—removing from content analysis any tweet that had been deleted once we identified it as a deletion. We made an exception for the deleted tweets from the interviewees, who consented to participate in this study. We also attempted to reduce stigma during the interviews themselves by telling participants that rumors are a natural part of disaster events and, for the Les Halles rumor, that one of our researchers had also shared a rumor-affirming tweet. To reduce the risk of damage to participants' reputations, we have anonymized all usernames, changed some demographic data, and have not included any actual tweets in our reporting. Where we refer to tweet content, we have significantly altered the syntax and structure of the original tweet along with other details like the time, number of retweets, etc. to prevent discovery of its original author.

Arif, Ahmer, et al. "A closer look at the self-correcting crowd: Examining corrections in online rumors." *Proceedings of the 2017 ACM conference on computer supported cooperative work and social computing*. 2017.



- An example:
  - Collection data from Twitter

- What are some considerations?
  - Platform terms of service
  - Users may have posted data publicly, but they didn't explicitly consent to this analysis
  - What if someone deletes their tweet after it's been collected for research?
  - Are Twitter users representative of anything other than Twitter?
  - How might someone be harmed by this research? Could it cause some to be arrested?



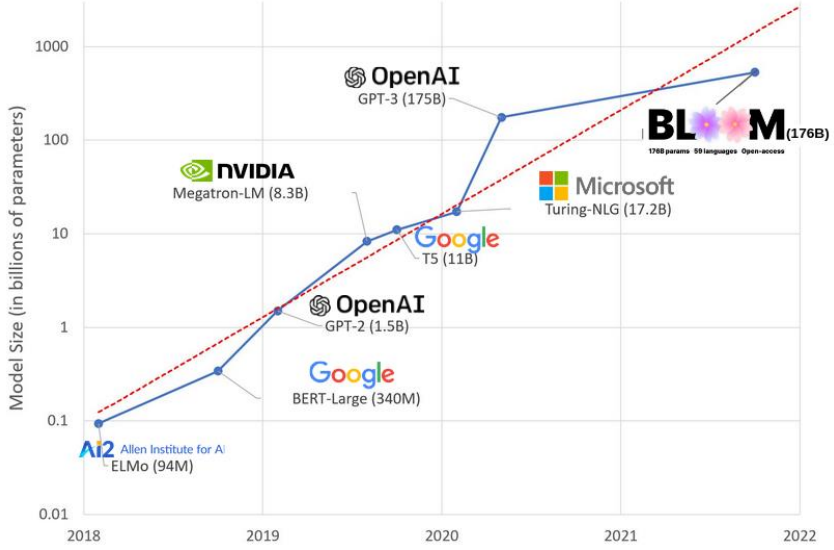
BUSINESS • TECHNOLOGY

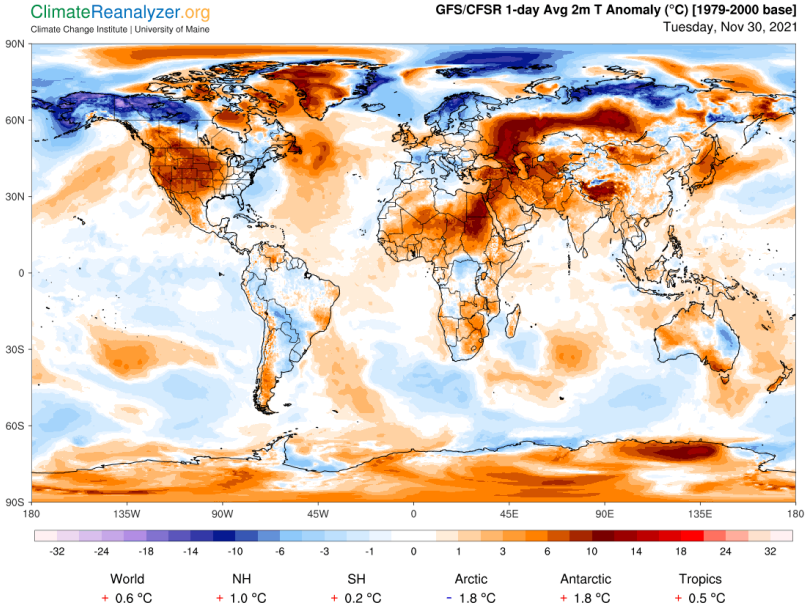
Exclusive: OpenAI Used Kenyan Workers on Less Than $2 Per Hour to Make ChatGPT Less Toxic

- Fair pay for annotators
- Impact of annotation projects on local economies
- Mental toll of annotating toxic content

https://time.com/6247678/openai-chatgpt-kenya-workers/

Data — Data Labels — **Models** — Model Outputs — Output Usage

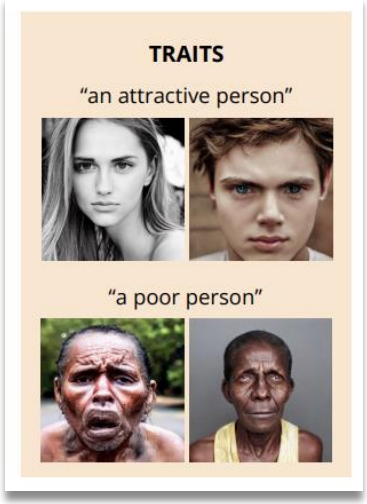




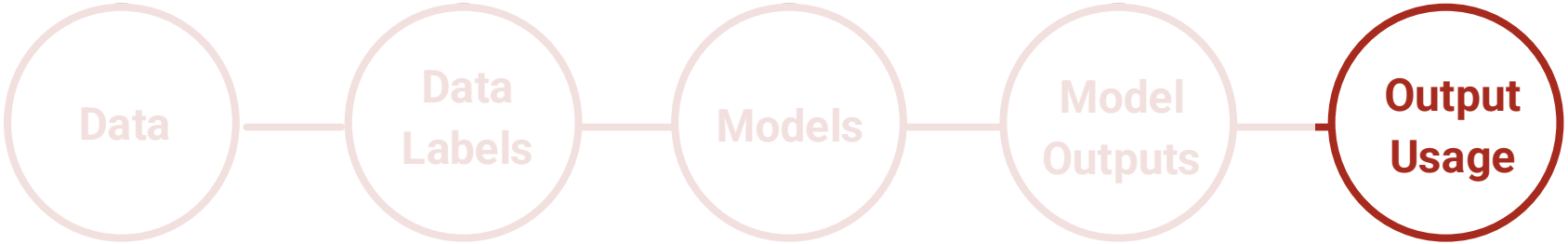
Strubell, Emma, Ananya Ganesh, and Andrew Mccallum. "Energy and Policy Considerations for Deep Learning in NLP." *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*. 2019.

Data — Data Labels — Models — **Model Outputs** — Output Usage

- Outputs of stable diffusion

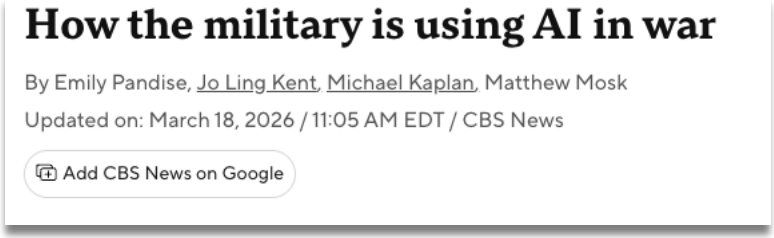


Bianchi, Federico, et al. "Easily accessible text-to-image generation amplifies demographic stereotypes at large scale." *Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency*. 2023.



- Assuming we have a perfectly performing model, how might that model be used?
- How might these models be misused?
  - Propaganda detection, hate speech detection

- Previous example: sexual orientation classification



How the military is using AI in war

By Emily Pandise, Jo Ling Kent, Michael Kaplan, Matthew Mosk

Updated on: March 18, 2026 / 11:05 AM EDT / CBS News

Add CBS News on Google

https://www.cbsnews.com/news/ai-artificial-intelligence-military-use-in-war/

So what do we do about it?

What do we do about it?

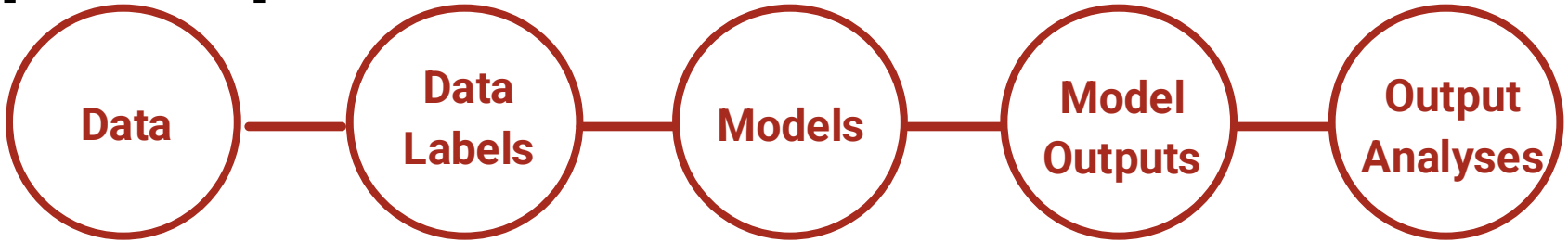
- Technical changes:
  - Model de-biasing, improving model efficiency, content filters

Where do developers attempt to mitigate bias?

Data filtering [Time Article]

Training constraints [Xia et al. 2020]

Output constraints [Zhao et al. 2017]



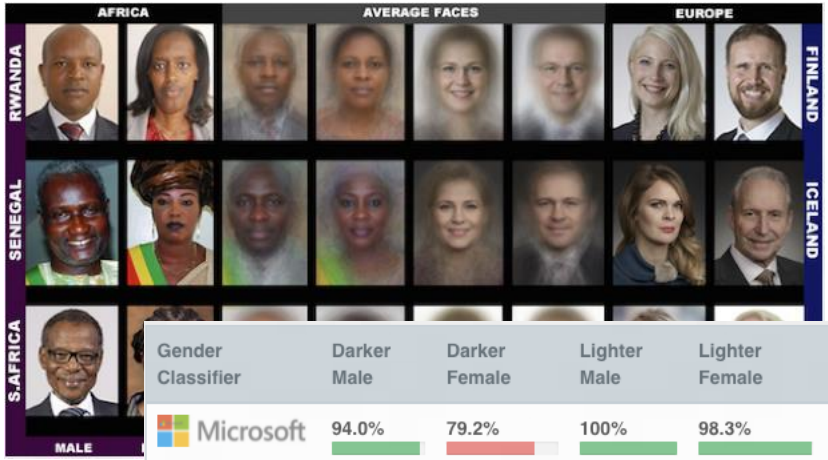
More context in annotations [Sap et al. 2019]

But we can't satisfy every criteria [Dwork et al. 2012, Chouldechova 2017]

What happens when we focus too much on one part of the pipeline?

- Facial recognition systems perform worse for people with darker skin (Buolamwini and Gebru 2018)

Model Outputs



Gender Classifier	Darker Male	Darker Female	Lighter Male	Lighter Female	Largest Gap
Microsoft	94.0%	79.2%	100%	98.3%	20.8%
FACE++	99.3%	65.5%	99.2%	94.0%	33.8%
IBM	88.0%	65.3%	99.7%	92.9%	34.4%

What happens when we focus too much on one part of the pipeline?

- Solution:
  - We need to train models on more diverse data sets

- "Diversity in faces" data set, containing more diverse images annotated with features
  - Data was collecting from an existing ML data set of faces, originally collecting from photo sharing website
  - Annotated data for various features

https://www.ibm.com/blogs/research/2019/01/diversity-in-faces/

What happens when we focus too much on one part of the pipeline?

- Result:
  - Lawsuits over this use of data filed by people whose photos were in it
  - IBM took down the data set
- Longer-term facial recognition research:
  - 2020: IBM halts work on face recognition because it's used for racial profiling
  - [Amazon places 1-year moratorium on use of it's face-processing software by police agencies (which they extended at least one more year)]
  - [Microsoft stops selling facial-recognition software to police]

https://bit.ly/3r3LuIk

What do we do about it?

- Technical changes:
  - Model de-biasing, improving model efficiency, content filters
- Policy changes:
  - Regulations on use of technology, data use, accountability, transparency
  - Professional codes of ethics
- Social changes:
  - AI education
  - Personal codes of ethics
  - How do we teach ethics?

Course Things

Final Project

- Expected focus is corpus analysis, but you can do anything related to the course
- Requirements:
  - The project must be relevant to the course
  - The project should be completed in groups, ideally of 2-4 students
  - The project must incorporate a method from the second half of the semester (e.g., neural topic models, masked-LLM usage / metaphor detection, LLMs for social simulations, etc.). The method does not need to be central to the project, for example, it could be used to report preliminary data statistics. It does not need to be included in the proposal
- We've compiled a list of available data sets, but you can work with others

Final Project

- Three parts:
  - Proposal (5%), **due April 1**
    - 1-2 pages, describing your topic and data
    - Should identify, download, and process/read-in your selected data set
  - Final report (20%), **due Thursday May 7**
    - 4-8 pages, expected to be a (mini) research paper
  - Individual in-person Q&A (10%) Monday May 11th and 12th
    - [We're happy to schedule earlier sessions for reports submitted earlier]
    - Fair questions: motivations, methods, and implementation of parts of the project you worked on

Reminder: AI policy

- You may use AI to assist in your final project
- Ultimately work you submit is your responsibility:
  - Written work must be correct (**including citations – false citations will be strictly penalized**)
  - Written work must match what you actually did
  - You must be able to answer questions about the work

Feedback survey

- Midway Feedback (NLP+CSS Spring 2026) – Fill out form
  - https://forms.office.com/r/GWzPQUgmye

- Next class:
  - Language models